# Analysing novel material properties using finite element analysis for prosthetic applications

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Shahmeer Satti
Islamabad, Pakistan
shahmeersatti18@gmail.com

Zartasha Mustanar
*School of Interdisciplinary Engineering and Sciences*
*National University of Sciences & Technology*
Islamabad, Pakistan
zmustanar@sines.nust.edu.pk

***Abstract*—This study is a brief analysis of 3 potential bone-substitute materials (BSM) via the usage of finite element analysis to determine their appropriateness and suitability as emerging candidates for use in future BSM applications.**

***Index Terms*—finite element analysis, bone substitute materials**

## I. Introduction

There is a shortage in the availability of autogenous and allogenous bone substitute materials (BSMs) for use in commercial medical applications despite their idealness for such uses, as evidenced by their high biocompatibility, osteoinduction, porosity, stability under stress, resorbability, degradability, plasticity, sterility and long-term strength and stability as implants [9]. This study aims to address the appropriate use of three notable, currently utilized alloplastic synthetic materials used as bone substitute materials in regards to their strength and mechanical properties when placed under large stresses by using a method of finite-element modelling to simulate such conditions.

## II. Background

The materials chosen for analysis and the reasoning for their selection is presented in this section.

### *A. Carbon Fiber/Poly Ether Ether Ketone (CF/PEEK)*

Polymeric composite materials are finding increased used as BSMs, however are restricted in their use by the needs for prosthetics in direct skeletal contact to have a low elastic modulus for structural compatibility but high strength. Purely polymeric materials are largely not appropriate for this purpose due to their low strength despite their low elastic modulus, therefore fiber-reinforced polymeric composites are more appropriate for this purpose due to their high strength and high adaptability by varying material arrangement and physical properties [12]. Carbon Fiber/Poly Ether Ether Ketone (CF/PEEK) was picked as an appropriate representative of this category of materials due to its biological inertness, compared to epoxy-reinforced polymeric composites which introduce concerns about toxicity, high mechanical strength, and high fatigue resistance in similar volumes to current metal implants [11]. The primary concern with the application of CF/PEEK as a material as a BSM relates to poor manufacturing methods producing poor quality CF/PEEK implants and prosthetics [7], however this is increasingly being resolved [6], making it an increasingly viable future candidate for use as a BSM in commercial applications due to its suitability for our requirements and relatively low cost to manufacture and produce.

### *B. Yttria-stabilized Tetragonal Zirconia Polycrystals (Y-TZP)*

Bioceramics have long found application and common widespread commercial utilization as BSMs across a variety of medical applications due to their versatility, common availability, desirable characteristics and relatively low cost. However, many would-be excellent candidates are restricted in their use due to mechanical limitations, lack of biocompatibility or lack of bioactivity [3]. Zirconia is an excellent candidate for this purpose, due to its excellent mechanical properties [2] and high biocompatibility, but insufficient bioactivity for use in certain applications [8]. However Yttria-stabilized Tetragonal Zirconia Polycrystals (Y-TZP) demonstrate superior characteristics in this regard to traditional Ceria-stabilized TZP as they undergo a lesser degree of low temperature aging degradation (LTAD), making them more suitable as BSMs for a wider variety of applications, which is noteable due to their excellent mechanical characteristics, namely some of the highest fractural toughness and flexural strength among bioceramic BSM candidates [5].

### *C. Ti-10Cr-20Nb*

The vast majority of orthopedic implants (approx. two-thirds) are made out of metals, making their commercial use and knowhow globally widespread. However, many metal alloys, namely common titanium alloys, cobalt-chromium and 316L stainless steel have relatively high elastic moduli (in the range of 100GPa to 210GPa) compared to bone, which normally has an elastic modulus of roughly 20GPa. Ti-10Cr-20Nb is interesting for the purpose that it has a similar elastic modulus to that of bone at 17GPa [13], and a significantly

higher yield strength and ultimate strength than currently commonly utilized metal alloys for orthopedic purposes. As a near -type alloy however, it demonstrates low mechanical strength at room temperature, which is not improved by current metallurgical processes such as heat treatment, however with future advances in metallurgy, derivatives of this alloy may become more suitable for orthopedic applications in the future, which is why this alloy in particular was selected for study [1].

## III. Methods

TABLE I
MATERIAL YOUNG MODULI, POISSON'S RATIOS AND DENSITIES

| Material | Young Modulus (GPa) | Poisson's Ratio | Density($\mathrm{kg\,m^{-3}}$) |
|---|---|---|---|
| CF/PEEK | 18 [10] | 0.350 [4] | 1400 |
| Y-TZP | 200 | 0.310 | 6070 |
| Ti-10Cr-20Nb | 17 [13] | 0.300 [14] | 5190 |
| Typical Cortical Bone [1] | 17 | 0.300 | 1850 |

Above we have detailed the values of the Young's Moduli, Poisson's Ratios and Densities utilized for the analytical models used in this study to determine the performance of each material under specified loan conditions that are detailed later onwards. We have utilized these properties in our application of finite element analysis (FEA) by applying a vertical compressive load of 500N on the femoral head of the femur to simulate the application of a load and compare relative performance to a typical cortical bone, included as a baseline and appropriately determine whether the selected materials present any promise as BSMs.

## IV. Results

TABLE II
SELECTED RESULTS UNDER AN APPLIED LOAD OF 500N TO THE FEMORAL HEAD OF A FEMUR

| Material | Von Mises Stress (Pa) | Total Deformation (m) | Equivalent Strain ($\mathrm{m\,m^{-1}}$) | Safety Factor |
|---|---|---|---|---|
| CF/PEEK | $1.386 \times 10^6$ | $3.140 \times 10^{-4}$ | $8.175 \times 10^{-5}$ | 15 |
| Y-TZP | $1.386 \times 10^6$ | $2.828 \times 10^{-5}$ | $7.363 \times 10^{-6}$ | 15 |
| Ti-10Cr-20Nb | $1.386 \times 10^6$ | $3.328 \times 10^{-4}$ | $8.664 \times 10^{-5}$ | 15 |
| Typical Cortical Bone | $1.386 \times 10^6$ | $3.328 \times 10^{-4}$ | $8.664 \times 10^{-5}$ | 15 |

Above we have listed results and below provided images achieved through FEA, including Von Mises Stress, total deformation under the selected load, equivalent strain, safety factors and a brief listing of key findings arising from the application of FEA under the stated conditions.

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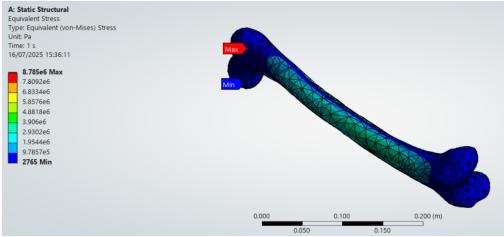


Fig. 1. CF/PEEK Von Mises Stress

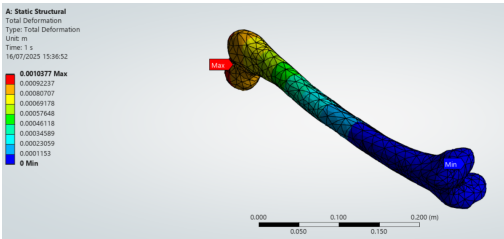


Fig. 2. CF/PEEK Total Deformation

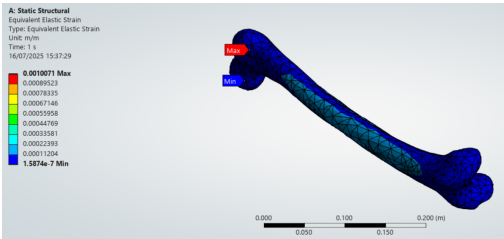


Fig. 3. CF/PEEK Equivalent Strain

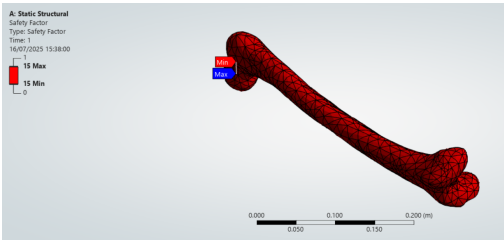


Fig. 4. CF/PEEK Safety Factor

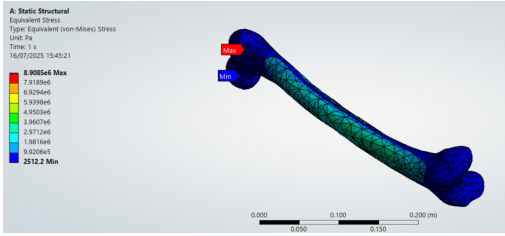


Fig. 5. Y-TZP Von Mises Stress

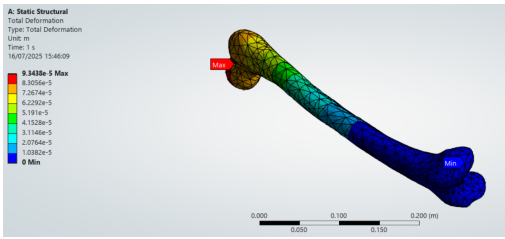


Fig. 6. Y-TZP Total Deformation

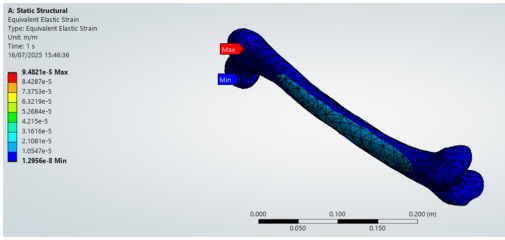


Fig. 7. Y-TZP Equivalent Strain

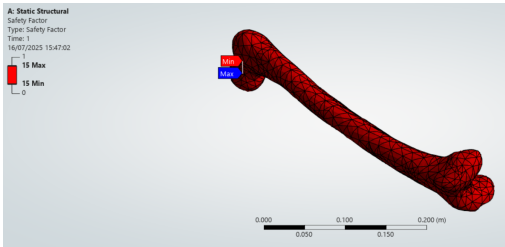


Fig. 8. Y-TZP Safety Factor

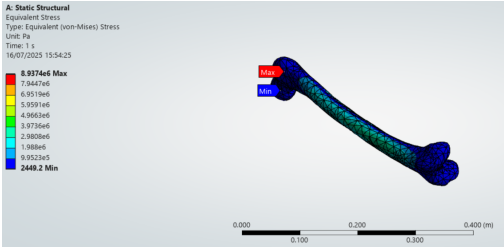


Fig. 9. Ti-10Cr-20Nb Von Mises Stress

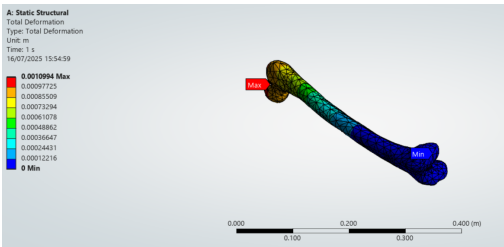


Fig. 10. Ti-10Cr-20Nb Total Deformation

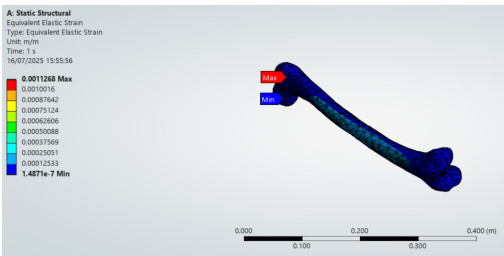


Fig. 11. Ti-10Cr-20Nb Equivalent Strain

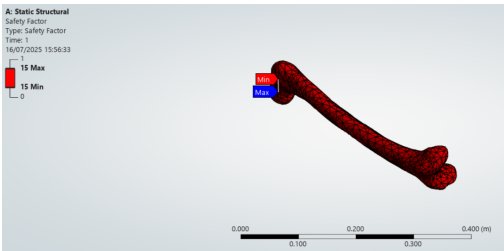


Fig. 12. Ti-10Cr-20Nb Safety Factor

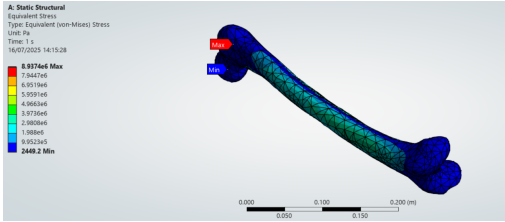


Fig. 13. Cortical Bone Von Mises Stress

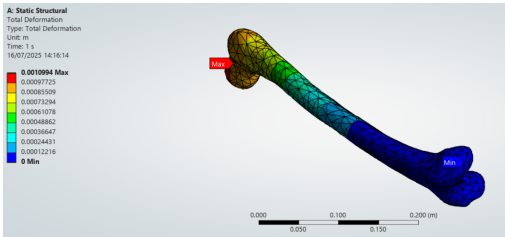


Fig. 14. Cortical Bone Total Deformation

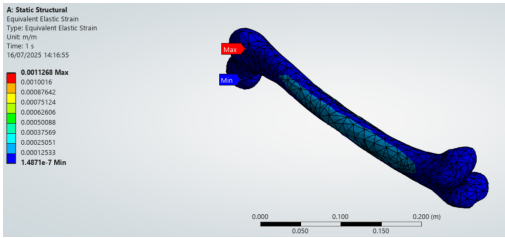


Fig. 15. Cortical Bone Equivalent Strain

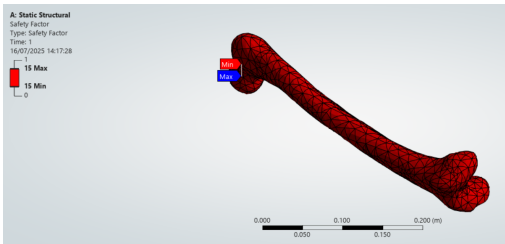


Fig. 16. Cortical Bone Safety Factor